

Pulkit Chawla

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SUMMARY

Software engineer (6+ years at Oracle Cloud and Qubole) turned AI engineer after an MSc in Data Science and AI at TU Eindhoven. In 2026 built and ran four agent systems of my own: a build gate that decides which AI-written code is allowed to ship, an AI that operates a Mac the way a person does, and an email agent I trust with my inbox. Owns services end to end: design, build, test, release, on-call.

Work authorisation: NL orientation-year permit (full labour-market access); HSM-eligible under the reduced graduate criterion

EXPERIENCE

Freelance and personal projects

Jan. 2026 – Present

Independent AI / software engineer

Eindhoven, Netherlands

- Built and run four agent systems (below): a build gate that decides which AI-written code is allowed to ship, an AI that operates a Mac the way a person does, a personal email agent I trust with my inbox, and a copy-trading platform that mirrors one order across many brokerage accounts
- Built a WhatsApp AI assistant for an Eindhoven student food-catering business (Twilio, Gemini/Gemma, FastAPI on DigitalOcean) handling orders, meal selection, leave days and billing questions

Eindhoven University of Technology, IMAGEN Programme

Jan. 2025 – Nov. 2025

Graduate AI Researcher (MSc thesis), MLOps and Computer Vision

Eindhoven, Netherlands

- Built the MLOps platform behind IMAGEN, a Dutch government-funded animal-welfare research programme run by TU/e, Wageningen University and Hendrix Genetics, on SURF's Spider HPC cluster and dCache storage
- Redesigned the pipeline from a PiCaS pilot-job setup into a decoupled control plane: ClearML experiment tracking and a CouchDB job queue feeding SLURM jobs that run in Singularity containers, with a web UI to configure clusters, experiments and runs
- Wrote the FAIR data pipeline that registers uploaded datasets, converts them to HDF5 and catalogs them with provenance; at eight loader workers HDF5 delivered 217.4 images/s against 112.3 for raw images, near-linear scaling that cut GPU idle time during training
- Exposed the pipeline through a web UI and REST API so researchers could fine-tune detection models without logging into the HPC cluster; per-animal bounding boxes fed behavioural analysis such as feeding-order dominance

Oracle Cloud Infrastructure

Jan. 2021 – Aug. 2023

Member of Technical Staff, Big Data Service (BDS)

Bangalore, India

- Sole owner of the customer-facing console of OCI Big Data Service in a ~30-engineer team: built, tested, released and operated the Node.js UI used to manage Hadoop clusters, and trained other engineers to take it over before leaving
- Owned the Hive and HBase distributions: forked upstream open source, set up TeamCity pipelines that built and deployed every PR to staging automatically, and kept both patched by back-porting upstream CVE fixes, a hard requirement for enterprise customers leaving Cloudera
- Served in the production on-call rotation, resolving P0/P1 incidents with the US-based support and architecture teams

Qubole (acquired by Idera)

Jan. 2018 – Dec. 2020

Software Engineer → Software Engineer II, Cluster Orchestration

Bangalore, India

- One of three engineers owning the cluster-orchestration backend of Qubole's multi-cloud big-data platform, spinning up and managing Hadoop, Spark, Presto and Hive clusters inside customer accounts on AWS, Azure, GCP and OCI
- Added Datadog monitoring for cluster metrics and Jenkins QA pipelines gating every release on client-specific integration tests across staging and production
- Built secure tunnels between Qubole's control plane and clusters running in customers' own cloud accounts, letting customers operate their clusters directly from the UI

PROJECTS & RESEARCH

devteam: a harness of microbots with contracts, for agent-written software |

Jul 2026 – present

Node.js

github.com/thispc/devteam

- My first version rotted: the agents kept editing their own code until every test passed and nobody could extend it afterwards. You cannot ask code to grade its own homework, so I moved the judging out of reach: the contract is hashed into the module's identity, the code runs in a container with no network, twice in two environments with the transcripts diffed, against tests it never saw

- I had built an excellent courtroom that never held a trial, so I ran one: gave an AI manager a brief and the power to hire its own team. It designed the crew, drafted contracts from three examples and shipped modules through the gate with no interventions from me. 356 commits
- Some things cannot be verified at all: network calls, clocks, human input. Those live in declared holes with an allowlist the owner grants, judged on recorded real traffic. A headless real-mouse probe layer caught fifteen bugs the unit tests had waved through

HumanHands: native macOS computer-use agent | *Swift, SwiftUI*

2026

github.com/thispc/humanhands

- An AI that operates a Mac the way a person does. It reads the screen through the accessibility tree rather than screenshots, so nothing leaves the machine, and it moves the real cursor along a human path instead of teleporting it, typing at human speed
- Passwords stay in the Keychain and reach the model only as placeholders, swapped in at the last instant. It refuses to type a secret until it has checked the site is the real one, and it can only touch the apps and sites I put on its list
- Fully native Swift across 10 SwiftPM targets with every OS primitive behind a protocol boundary; 135 tests run off-device

Personal email agent (NanoClaw fork) | *TypeScript, Node.js, Claude Agent SDK*

2026

github.com/thispc/nanoclauw

- An agent that reads my email can be talked into things by anyone who emails me, so the model that touches raw mail has no tools at all. The agent that acts sees only a cleaned-up summary, and anything going out stops at a one-tap approval that sends exactly the draft I saw, enforced in code rather than in a prompt
- Extended the open-source NanoClaw runtime with an iMessage channel and a portable plain-file knowledge base (+16k lines); runs 24/7 on own hardware across three daemons (iMessage, ingestion, PWA) at zero idle cost

jobhunt: agent-operated job-application system | *Python, FastAPI, Playwright*

Sep 2026

- Directed and shipped, with Claude Code, a pipeline that polls nine applicant-tracking systems across 160+ Dutch companies, matches employers against the 12,900-row IND sponsor register, scores postings, tailors a LaTeX CV per job from an evidence-tagged profile, and fills application forms with Playwright behind a submit guard

LLM question answering with RAG (TU/e summer research) | *Python, LangChain*

Jun – Aug 2024

- Built a question-answering pipeline over a corpus of 10,000+ academic documents using LangChain, the GPT-4 API and FAISS vector search with OpenAI embeddings, keeping retrieval under 100 ms per query
- Benchmarked against BM25 and TF-IDF baselines and tuned chunking, context management and prompting (chain-of-thought, few-shot) to beat plain LLM answers on domain-specific queries

EDUCATION

Eindhoven University of Technology (TU/e)

Aug. 2023 – Nov. 2025

Master of Science in Data Science and Artificial Intelligence

Eindhoven, Netherlands

- Thesis: A Reimagined and Implemented MLOps Architecture for Supercomputing Environments (IMAGEN programme, SURF Spider HPC), supervisors Dr. Yanja Dajsuren and Manu Agarwal

Birla Institute of Technology, Mesra

Aug. 2014 – Jun. 2018

Bachelor of Engineering in Information Technology, CGPA 8.37/10, First Class with Distinction

Ranchi, India

- CGPA 8.37/10, First Class with Distinction

TECHNICAL SKILLS

AI/ML: LLM agents (Claude Agent SDK), RAG and vector search (FAISS), prompt-injection defences, LangChain, PyTorch, computer vision (object detection, tracking), fine-tuning, model serving and inference services, ClearML

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, Bash

Backend & Tooling: FastAPI, Spring, Node.js, REST APIs, Postgres, Redis, Jenkins, GitHub Actions, Ansible, CI/CD, Datadog, Elasticsearch, Playwright

Big Data & Cloud: Hadoop, Spark, AWS, Azure, OCI, GCP, Docker, Kubernetes, Ansible, Postgres, Redis

Spoken languages: English: fluent; Hindi: native; Dutch: beginner, learning